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Cross-Dataset Generalization for Network Intrusion Detection Using Supervised and Unsupervised Machine Learning Models

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Abstract

Network Intrusion has been on the rise ever since the invention of internet and IoT devices that specifically accumulate huge amounts of data. Network Intrusion Detection System (NIDS) has been a major topic considering the number of threats involved in today's cyber security scenario. With the ever-rising threats in communication networks, development of NIDS plays a boosting factor to prevent the networks from attacks of all kinds. The purpose of the development of network intrusion detection system is to protect the communication network, preserve and safeguard vital information from intruders dynamically by analyzing the complex and sophisticated traffic data. The motives for the intrusion vary on individual and institutional levels, financial gains, unauthorized access of intellectual properties, confidential information, political gains and others. NIDS monitors and analyzes all the incoming web traffics, detects them as 'Attack' or 'Benign' thus safeguarding the network from future attacks of same kinds. Several research has been carried out to detect network intrusion for specific fields (Health, Finance, Entertainment, Military, Government) incorporating advanced techniques with the utilization of specific data. Previously published research papers showed that the deployment of robust machine learning algorithm achieved greater performance in intrusion detection when trained and tested on the same dataset. Research portals such as JSRT Journal, MDPI, ResearchGate, ScienceDirect, Springer Nature, IEEE contain research papers related to cross dataset generalization (mostly between CICIDS2017 and CSE-CIC-IDS2018). Most of the research papers drew their conclusions based on 'Accuracy' of the models for intrusion detection. There was no uniformity in these papers. Some described the usage of SMOTE while training CICIDS2017 dataset in supervised learning models, variation on the size of dataset and percentage of data used for training the models; thus, leaving a gap for cross-dataset generalization to the unseen network traffic. This research paper tries to showcase the capabilities of Supervised Learning Models (Random Forest, XGBoost), Unsupervised Learning Models (PCA, K-Means, Autoencoders) and Deep Learning Models in cross-dataset generalization. All the models will be trained on CICIDS2017 dataset and tested on two different test datasets namely CSE-CIC-IDS2018 and UNSW-NB15; which are publicly available. No SMOTE will be applied to oversample the CICIDS2017 dataset. Unlike previous papers, this research will prioritize the evaluation metrics significant for cross-dataset generalization. Finally, the performance of the models will allow us to assess the strength and practicality of a unique approach in real-world cybersecurity scenario.

Declaration of Originality

We confirm that our thesis "Cross-Dataset Generalization for Network Intrusion Detection Using Supervised and Unsupervised Machine Learning Models" presented here is original and has been submitted to the Gisma University of Applied Sciences to fulfil the requirements for the degree of MSc Data Science, AI and Digital Business.

We assure through our awareness that the thesi with this topic hasn't been presented to any other academic institution or university for the award of the academic degree. We assure that the thesis has been completed through our collective effort under the guidance of our supervisor incorporating approved guidelines and regulations, acknowledgement and citation of all resources, data and materials. Output analysis, interpretations and conclusion drawing were done with active supervision of our supervisor. Any contributions of authors have been duly recognized and cited.

We are fully responsible for the originality and integrity of the contents written in the thesis.

For the convenience of academic evaluation, we have uploaded the necessary resources at '<https://github.com/9sumit9/Dissertation-MSc-Data-Science-AI-and-Digital-Business>'.

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Dedication

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Chapter 1

Introduction

The growth of internet has significantly assisted humans, organizations and government in ways no one ever dreamt of ages ago. Internet has simplified the lives of individuals and organizations. Development of robust communication platforms, enhanced economy across the globe, elevation in military operations, education, health, entertainment, science, research and invention are some of the privileges provided by the internet. Along with the privileges come the burdens in the form of cyberattacks. Intruders tend to overcome the traditional security firewalls and benefit by constantly bombarding malicious attacks that the obsolete web security tools and even antiviruses software like Kaspersky could never detect [Miani et al., 2025]. Thus, urges the need for the deployment of Network Intrusion Detection System (NIDS). Robust AI mechanism and machine learning algorithms assist NIDS to detect sophisticated unseen attacks. They help in identifying the patterns of the malicious attacks analyze them and blocks them enabling a secured network infrastructure in the future. The deployment of Supervised Machine Learning models such as Random Forest, XGBoost, Unsupervised Learning models (Isolation Forest, One-Class SVM, Autoencoders) and Deep Learning models exhibit better performance in intrusion detection from unknown attacks with improved accuracy and F1- Score [de Carvalho Bertoli et al., 2023]; [Hamidou and Mehdi, 2025]; [Wicaksana and Umam, 2026] particularly on CICIDS2017 and UNSW-NB15 datasets easily available on public domain. Unlike previous datasets such as KDD Cup 1999 and NSL – KDD, these datasets contain real network traffic with varied attacks inhibiting wider traffic patterns [Moustafa and Slay, 2015, Sharafaldin et al., 2018]. Even though several AI driven approaches have been considered and developed to detect intrusion in the computer network with better accuracy, some models have notable performance on the benchmark datasets (CICIDS2017, CSE-CIC-IDS2018 and UNSW-NB15) alone but fail while deployed to detect unknown incoming traffic and thus underperform considering their capabilities in cross-dataset generalization across varied computer network. Intruders take an upper hand in this scenario by deploying more sophisticated versions of web attacks continuously [ul Haq Qazi et al., 2022]. Some of the reasons why these models fail to generalize conveniently.

- Class Imbalance – Most of the web traffics are ‘Genuine’ and few are actual ‘Attack’.
- Variable features and size – The training and testing datasets consist of different features; also, the size of the features sourced from different computer network alter.
- Computational power – Researchers at an individual level often find it difficult to evaluate the performance of the models given the size of the datasets and the computational power of the device.

These issues motivate the researchers to find alternative ways to build robust NIDS to safeguard the computer network from intruders. This allows researchers to learn new patterns of attacks and dynamically adjust the models from a classical approach to a revolutionary approach [Kottilingal, 2024].

1.1 Underpinning Theory

1.1.1 Anomaly-Detection Theory

Anomaly-detection theory provides the principal foundation for this research. It assumes that normal observations exhibit an underlying structure that can be learned and that observation departing sufficiently from this structure may be anomalous. In flow-based network intrusion detection, an observation represents a network flow described by statistical measurements such as duration, packet count, transferred bytes, mean packet lengths, and traffic rates. A detector learns the expected distribution or structure of these characteristics and assigns higher anomaly scores to observation that depart from the learned representation.

Unsupervised anomaly detection reduces dependence on labelled attack examples. This is useful because attack labels can be expensive to produce, incomplete, inaccurate, or unavailable for emerging threats. Models trained primarily on normal traffic may also identify previously unseen attacks because they are not restricted to recognizing attack classes observed during training. However, unsupervised detection does not automatically establish that an unusual observation is malicious. Changes in legitimate applications, devices, users, and traffic volumes may also appear anomalous. Similarly, attacks designed to imitate ordinary behavior may remain close to learned normal representation [Pang et al., 2021, Chou and Jiang, 2022]. Anomaly detection therefore involves uncertainty and should not be interpreted as a perfect separation between benign and malicious traffic. Operational performance depends on the quality and representativeness of the normal data training data, the selected features, the anomaly-scoring mechanisms, and the decision threshold. The anomaly-detection process contains two distinct decisions. First, a model produces a continuous anomaly score. Secondly, an operating threshold converts that score into a benign or attack prediction. A model may successfully rank attack observations above benign observations while producing weak binary classifications if the selected threshold is inappropriate. This distinction is central to the interpretation of ROC-AUC and F1.Score in the present study. ROC-AUC evaluates the ranking of continuous anomaly score across possible thresholds. F1.Score evaluates the binary classifications produced using one selected operating threshold. Reporting both measures helps distinguish weak anomaly representation from weak threshold transfer.

1.1.2 One-Class Learning

One-class learning builds a representation of one class rather than learning decisions boundaries among several labelled classes. In anomaly-based network intrusion detection, the model is commonly fitted using benign observation and treats departure from the learned profile as potential attacks.

This approach reflects the practical asymmetry between relatively abundant normal traffic and incomplete knowledge of possible attacks. It may therefore provide a mechanism for identifying attacks that were not represented during model training.

The validity of one-class learning depends on several assumptions. Normal training data must be sufficiently representative substantial contamination must be absent, and the selected features must preserve meaningful behavioral differences.

If attack observations contaminate the normal training partition, the model may learn some malicious behavior as normal. Conversely, if the normal training data represent only a narrow range of legitimate behavior, the detector may incorrectly classify unfamiliar benign traffic as malicious [Pang et al., 2021, Yang et al., 2022].

The unsupervised models evaluated in this research implement one-class learning differently.

Isolation Forest identifies observations that are easier to isolate than normal observations. One-Class SVM estimates a boundary enclosing normal traffic. The improved autoencoder learns a reconstruction function from normal observations.

Comparing these mechanisms under a consistent experimental protocol helps determine whether any particular representation of normality is more transferable across independently collected datasets.

1.1.3 Dataset Shift and Domain Shift

Dataset shift occurs when the statistical distributions encountered during testing differs from the distributions used during training. Domain shift is related concepts describing differences between the source environment used for model development and the target environment in which the is evaluated or deployed.

In network intrusion detection, these differences may arise from:

- network architecture and topology;
- available services and applications;
- communication protocols;
- user and device behavior;
- traffic volume and timing;
- attack implementation;
- capture tools and flow exporters;
- feature definitions;
- sampling procedures; and
- class proportions.

Dataset shift is therefore not an isolated defect. It represents a collection of differences that can affect both benign and malicious traffic. A model trained using one network may interpret legitimate traffic from another network as anomalous because its applications, devices, or traffic volumes differ.

Within-datasets evaluation may conceal these problems because the training and test partitions share a common origin. Even when individual observations do not overlap, both partitions may retain the same traffic-generation process, feature distributions , and collection artefacts.

High within-dataset performances may consequently represent interpolation within one benchmark rather than transfer to an independent environment [Kenyon et al., 2020, Goldschmidt and Chudá, 2020].

Cross-dataset evaluations provide a stronger test because model training and testing use independently collected datasets. [Verkerken et al., 2022] evaluated an autoencoder, Isolation Forest, One-Class SVM, and Principal Component Analysis using CICIDS2017 and CSE-CIC-IDS2018.

Although the models performed strongly during individual-dataset evaluations, their average ROC-AUC decreased by approximately 30.45% under inter-dataset evaluation. This finding demonstrates the gap that can exist between internal performance and external generalizations.

[Cantone et al., 2024] similarly found that models obtaining strong within-dataset results could perform close to chance in cross-dataset settings. Their analysis associated weak transfer with dataset heterogeneity and differences in underlying features distributions.

Together, these studies support the central theoretical expectations of the present dissertation: cross-dataset evaluation is expected to produce lower performance than within-dataset evaluation. However, the extent of the deterioration may depend on the model, source dataset, target dataset, and direction of transfer.

1.1.4 Generalization Theory

Generalization refers to model’s ability to perform effectively on observations that were not used during model development. Cross-dataset generalizations imposes a stronger requirement because the unseen observations originate from an independent data collection rather than a held-out partition of the source dataset.

A transferable intrusion detector should learn relationships that remain useful when dataset characteristics change. It should not depend primarily on collection artefacts, dataset-specific feature ranges, or attack patterns found only in the source dataset. A reliable cross-dataset experiment must prevent target-domain information from influencing preprocessing, model fitting, model selection, or threshold calibration. Fitting a scaler using combined source and target data, selecting hyperparameters based on target performance, or maximizing target F1_Score would introduce information leakage.

The resulting performance would describe target-assisted adaptation rather than strict external generalization. Transparent data handling and clear separation of source and datasets are therefore essential [Ring et al., 2019, Yang et al., 2022, Goldschmidt and Chudá, 2025].

In the present research, anomaly representation are fitted using normal source-domain traffic. Labels from a held-out source validation partition support model selection and decision-threshold calibration. The procedure is therefore described as *unsupervised representation learning with source-domain validation calibration* rather than as a completely label-free pipeline. Target labels are reserved exclusively for the final calculation of evaluation measures.

Chapter 2

Literature Review

Some of the research carried out on Cross-Dataset Generalization for NIDS have shown improvements with advanced dataset handling and feature engineering on supervised machine learning and deep learning models. These models have achieved greater accuracy on benchmark datasets and improved to some extent in cross-dataset generalization (between CICIDS2017 and CSE-CIC-IDS2018, UNSW-NB15). However, accuracy alone cannot be considered to determine how robust the model is given the high number of genuine network flows and a smaller number of attacks. Cross-Dataset Generalization for NIDS face reliability issue when models are deployed for intrusion detection considering the increasing number of attacks in heterogeneous environments with huge domain shifts when the models with greater accuracy on benchmark dataset fail to detect intrusion in an unknown dataset yielding more False Positives. Generalization becomes complex as the features from the train dataset that are predictive of malicious activities within the dataset may fail to predict malicious behavior from a different dataset thus dropping the performance of the models [Elangovan et al., 2026]. Uniformity in dataset size for training and testing, usage of SMOTE are crucial to make comparisons on how the models performed in intrusion detection in the future although they vary on design, features and intended research goals. Recording environments, data volume, realism, evaluation design, anonymization and attack distribution have their limitations, thus prioritizing the selection of datasets based on research goal rather than just popularity [Ring et al., 2019, Thakkar and Lohiya, 2020, Kenyon et al., 2020]. Moreover, popular datasets only include a part of the current threat environment which seem to be biased in detection performance [Hindy et al., 2020]. A reliable NIDS evaluation depends on the experimental pipeline, including preprocessing, feature generation, detection methodologies and evaluation metrics rather than just the model selection [Yang et al., 2022]. The common issue with data quality and applicability is the usage of CICIDS2017 as the benchmark dataset [Goldschmidt and Chudá, 2025]. Similar features indicate distinct underlying properties across datasets due to variations in traffic capture, dataset production and feature construction [Pinto et al., 2025].

2.1 Supervised Machine Learning

The studies have shown the effectiveness of ensemble learning and feature selection in finding nonlinear dependencies among different labels in the network flow by keeping low dimensionality. They have also cautioned to interpret the evaluation wisely despite Random Forest, XGBoost, 1D-CNN achieving accuracy 85% on benchmark datasets [Elangovan et al., 2026], these models faced challenges modelling on complex multi-stage attacks [Chandra and Anandappa, 2025].

With the ever-increasing study and experiments on network intrusion detection, a recent study by [Cantone et al., 2024], although not validated and reviewed by experts performed cross-dataset generalization by employing machine learning algorithms on CIC-IDS-2017, CSE-CIC-IDS2018, LycoS-IDS2017, and LycoS-UnicasIDS2018 datasets and eventually applied corrections to CSE-CIC-IDS2018 based on the results of LycoS-IDS2017. The study revealed that the performance dropped to detect malicious behaviors for generalization in comparison to near perfect results in intrusion detection when trained and tested on same datasets. The supervised learning models learned bigger scale attack types such as DOS and DDoS and generalized well.

Meanwhile, due to class imbalance of the attack types, the models failed to detect web attacks and infiltration attacks. The researchers have pointed out that the variability of the datasets, irregular distribution of attacks among datasets and redundancy in the samples played a causal role in performance drop of the models failing to learn the attack patterns. Problems associated with feature calculation, protocol identification, feature multiplication and labeling in CIC-IDS-2017 degraded the quality of the data and hence corrected with LycoS-IDS2017. The findings suggested that the learning of the transferrable attacks might not be the real reason, but the features associated with the dataset that hinder the performance of the models. As per the study, the models should be implemented with prior knowledge of different traffics that were never encountered before.

The table below showcases the performance of the models while trained on CIC-IDS-2017 and tested on CSE-CIC-IDS2018 as per [Cantone et al., 2024] for cross-dataset generalization for network intrusion detection.

Table 1: Model Evaluation Results from [Cantone et al., 2024].

Train Dataset	Test Dataset	Models	Evaluation Metrics		
			MCC	F1	ROC-AUC
CIC-IDS-2017	CSE-CIC-IDS2018	Random Forest	25.96%	25.76%	66.93%
		XGBoost	35.68%	26.43%	80.44%

Another study by [Elangovan et al., 2026] revealed the performance drop for NIDS while evaluated on recent datasets. Random Forest when trained on CIC-IDS-2017 and tested on CSE-CIC-IDS2018 saw a Macro F1 drop of 0.059 from baseline 0.941 to 0.882. Similarly, 1D-CNN saw a decline of 0.070 from 0.942 to 0.872. It seems obvious that the performance degrades while generalizing to a completely different dataset when both temporal and network domain distance increases implying that performance on benchmark dataset does not signify robustness for future attack detection. The study highlights the importance of carrying out temporal forecasting evaluation while implementing models for intrusion detection.

2.2 Hybrid Deep Learning approach for Cross-Dataset Generalization

In addition to the application of supervised learning models, usage of deep learning methods has been observed for the detection of intrusion due to its capability of learning hierarchical representations from the complex traffic data. A research paper “Real-Time Network Intrusion Detection System Using Deep Learning: Identifying Cyber Attacks Through Anomaly Pattern Recognition” [Chandra and Anandappa, 2025] has properly analyzed the performance of LSTM, CNN and Autoencoders in detecting anomalies. The paper has discussed the hybrid approach and proposed the Real-Time NIDS by employing attention mechanism, LSTM, CNN and Autoencoders. In this hybrid architecture setup, LSTMs model network flow sequences, CNNs identify spatial patterns from the traffic features, Autoencoder calculates the anomaly score from the reconstruction error and lastly the important characteristics/temporal phases within the traffic data sequence are identified by attention mechanism.

Like the performance of the supervised learning models on benchmark datasets, application of deep learning for intrusion detection recorded exemplary F1 scores of 97.8%, 97.3% and 98.0% for CIC-IDS-2017, CSE-CIC-IDS2018 and UNSW-NB15 dataset respectively. The study has conducted cross-dataset performance between CIC-IDS-2017 and UNSW-NB15 only. Using this hybrid approach, they have yielded a F1-score of 93.5% and FPR of 2.8% only which has surpassed the performance of the base supervised learning models. The major difference in the

results between the application of supervised learning models from previous research and the hybrid DNN approach lie in data split for training and testing. The study of cross-dataset generalization by [Cantone et al., 2024] used whole dataset for training and testing. Meanwhile, the research paper by [Chandra and Anandappa, 2025] used SMOTE (to oversample rare attack classes), 70% of each dataset for training, 15% for hyperparameter tuning and remaining 15% for testing. While this approach yielded superior performance, as per Elangovan et al. (2026), this complex deep learning architecture does not guarantee a reliable generalization as the performance for intrusion detection degrade when there are large distribution shifts and may not be resilient enough to identify unfamiliar traffic thus contradicting the notion that complex model would yield better performance when the models are trained on older traffics and tested on newer version of web traffics with evolving network technologies, communication behaviors and varied attack patterns. Despite the robustness of the models, [Elangovan et al., 2026] has stated the necessity of preserving the network features when the network traffic features are classified based on their semantics during dataset shift. Protocol specific indicators showing higher degree of uncertainty whereas communication features like flow rate and temporal activity features generalizing well on different datasets. The negative communication behavior aspects that arise in different situations can be handled by certain behavioral characteristics. The evidence for this is discussed in the representation-drift study by [Elangovan et al., 2026]. Higher the difference between training and testing datasets with respect to time and field, higher the construction error. Models trained on CIC-IDS-2017 showed 0.0211 mean reconstruction error while analyzed on CSE-CIC-IDS2018, increased to 0.0342 for CICIoT2023 and reached the maximum of 0.0384 when applied to CIC-DIAD2024 showing the latent representations obtained from the original dataset diverged from the target-domain traffic.

The findings from the research for cross-dataset generalization by [Chandra and Anandappa, 2025, Elangovan et al., 2026, Cantone et al., 2024] are in contrast considering the models used. These papers show how the evaluation design, datasets, pre-processing, feature engineering and model design affect generalization. Thus, cross-dataset generalization for NIDS remains continuous deep research of the specific source and target datasets, features compatibility, attack overlap and experimental approach.

2.3 Unsupervised Machine Learning

Unsupervised anomaly detection is attractive because it reduces dependence on comprehensive attack labels. It attempts to learn regularities in normal traffic and identify observations that depart from those regularities [Pang et al., 2021, Chou and Jiang, 2022].

Isolations Forest is suitable for large tabular datasets because it use random partitioning rather than estimating a complete probability distribution. However, its definition of isolation remains connected to the source distribution. Normal target observations outside the source features ranges may therefore receive high anomaly scores.

One-Class SVM constructs a flexible non-linear boundary around normal observations. However, its performance is sensitive to feature scaling, kernel parameters, and training sample size. This sensitivity becomes particularly important when features distribution change between datasets.

Autoencoders learn non-linear representation of normal traffic and use reconstructions error as an anomaly score. Their additional capacity support the modelling of complex relationships among anomalous observation accurately and reduce the separation between benign and attack traffic.

Regularisation, early stopping, architecture selection, and careful threshold calibration are

therefore required [Pang et al., 2021, Yang et al., 2022]. These design choice may reduce over-fitting, but they do not guarantee cross-dataset generalisation.

No model is theoretically guraranteed to generalise best across all dataset and transfer directions. Their relative performance must therefore be determined using an identical experimental protocol.

[Alhajjar et al., 2021] demonstrated that machine-learning-based network intrusion detectors are vulnerable to adversarial examples. This finding shows why high benchmarks performance should not automatically be interpreted as evidence of operational security.

The threat and dataset analysis presented by [Hindy et al., 2020] similarly demonstrated that limited threat coverage restricts the conclusion drawn from benchmark. A model cannot demonstrate reliable detection of attacks that are absent or inadequately represented in the evaluation data.

The present dissertation does not conduct an adversarial-evasion experiment. However, strict cross-dataset evaluation provided a relevant test of robustness by exposing each detector to independent network distributions that were not used during model development. The study closely related to the present methodology is [Verkerken et al., 2022]. The authors compared an autoencoder, Isolation Forest, One-Class SVM using CICIDS2017 and CSE-CIC-IDS2018. Their results demonstrated that strong individual-datasets performance did not imply successful inter-dataset transfer. Average ROC-AUC decreased by approximately 30.45% when models were evaluated using traffic from a different dataset.

The principal cross-dataset setting used by [Verkerken et al., 2022] involved two related CIC datasets. The present dissertation extends this scope by including UNSW-NB15, which was generated using a different network environment and feature schema. [Cantone et al., 2024] provided further evidence that within-dataset success may conceal weak external performance. Although their models and complete dataset selections do not match those used in this dissertation, their findings reinforce the need to evaluate multiple directional transfer paths and examine dataset heterogeneity.

Published studies frequently use different features, data partitions, class proportions, threshold-selection procedures, and metric definitions. Their reported values therefore contextual benchmark rather than directly equivalent observations claims that one study outperformed another should only be made when the experimental conditions are sufficiently comparable.

The present research evaluates three within-datasets paths and six cross-dataset directional among CICIDS2017, CSE-CIC-IDS2018 and UNSW-NB15 datasets. This design supports comparison between transfer involving the two CIC datasets and transfer involving the more structural distinct UNSW-NB15 datasets

Federated learning provided a mechanism for a collaborative model development without centralizing raw network records. [Gourceyraud et al., 2025] Proposed and supervise federated intrusion detection architecture using clustering and federation initialization. Their research connect unsupervised learning, privacy prevention and multi-source data. [?] Propose supervise federated learning approach for intrusion detection across heterogenous networks. Their study further demonstrate the relevance of multi-source learning to the generalization problem.

However, these studies does not provide directly comparable baseline for three models and six cross data parts evaluated in this thesis. They are therefore used to motivate further research rather than to claim that the present models outperformed federated architectures.

2.4 Research Gap

All these previous studies of Cross-Dataset generalization for NIDS using Supervised Machine Learning and Deep Learning Models demonstrate gradual progress for NIDS, although the evaluations were made using different metrics on different test datasets. Trees based models seem to provide faster classification benchmarks while deep learning methods contribute to representation learning, temporal modelling, and anomaly detection. Hybrid deep learning method could significantly assist real-time cross-dataset generalization for NIDS. However, achieving accuracy greater than 80% or even 90% with high precision scores on a single benchmark dataset or a cross-dataset (as the results suggested by the studies above) does not confirm its robustness for deployment. They fail to exhibit excellent generalization capabilities. There is no consistent performance across varied test datasets thus failing to deliver similar results. The inconsistency in performance across datasets proves the significance of good labeling, dataset preparation and feature engineering. There remains a consistency gap in the advancement of NIDS evaluation approaches that would consider the efficiency of classification and generalization to different datasets, robustness over time, feature stability, imbalance class, false-positive estimations, and computational efficiency. Future research should conduct collaborative experiments for real deployment simulations with independent datasets and heterogeneous network conditions.

The literature review also establishes four principal findings. First, unsupervised anomaly detection reduces dependency on the complete attacks labels. this property makes it relevant to a margin attacks for which reliable labels are available. Secondly isolation forest one class as vm and auto encoder use materially different representation of normal traffic. the assumptions may cause them to respond differently to network traffic distribution shift. Third, data construction, future engineering, preprocessing, threshold selection and metrics definition substantially influence reported performance. High results cannot be attributed to the model without considering the complete evaluation pipeline. Fourth, strong within data result do not demonstrate external generalization. model may exploit characteristic specific to one dataset and fail when applied to independently collected traffic. Despite this evidence, a research gap remain in consistent , fully directional evaluation of the three selected model across the C17C18NB15. The present desertion address this gap through.

- a common nine-feature representation
- source only data cleaning and pre processing
- normal only source domain model fitting
- explicit source validation model selection
- source validation threshold validation
- three within data baselines
- six directional cross data paths
- no data domain restraining or threshold recalibration
- consistent performance measures

This design provides a consistent basis for measuring and comparing the cross data robustness of Isolation Forest One Class SVM and improved autoencoder.

2.5 Research Questions

To serve as a guide for this research overcoming the research gaps discussed earlier, we have developed the 'Qualitative' research questions that specifically contribute towards the efficiency of both supervised and unsupervised machine learning models for network intrusion detection on different datasets.

The following are the qualitative research questions developed for the research:

1. How well do supervised and unsupervised machine learning models trained on CIC-IDS-2017 dataset generalize to unseen intrusion detection datasets?
2. How does the cross-dataset generalization performance of unsupervised models compare with supervised models?
3. Which machine learning model demonstrate greater robustness against dataset distribution shift in real-world network intrusion detection scenarios?

2.6 Research Hypotheses

Cross-dataset generalization for network intrusion detection is crucial in determining the robustness of the models in identifying intrusion from a varied unknown network traffic. As previous studies have stated that although the models demonstrated exemplary performance while trained and tested on same models; the models saw significant drop in performance and struggled to generalize efficiently with respect to unseen dataset environment comprising of varied traffic characteristics and attack types. Based on these facts, we have formulated few hypotheses to investigate the cross-dataset generalization capabilities of the supervised and unsupervised learning models for developing a robust network intrusion detection system.

- **H1:** The evaluation metrics Precision, Recall, F1_Score and ROC-AUC for intrusion detection depends significantly on the types of model used.
- **H2:** The evaluation metrics of the models for intrusion detection depends significantly on the assessment setting used for evaluation.
- **H3:** Attack type signifies the performance of the models in terms of evaluation metrics.
- **H4:** Cross-generalization for network intrusion detection effectiveness relies on the interaction between both the selection of models and evaluation settings.
- **H5:** Robustness is not determined solely by the highest score on one transfer path.
- **H6:** Evaluation setting and attack type has significance in generalization for intrusion detection.
- **H7:** Dataset features have intermediary role between the individual variables and dependent variables for the effectiveness of intrusion detection.
- **H8:** Feature selection/types mediates the link between the model and the performance of intrusion detection.

2.7 Conceptual Framework

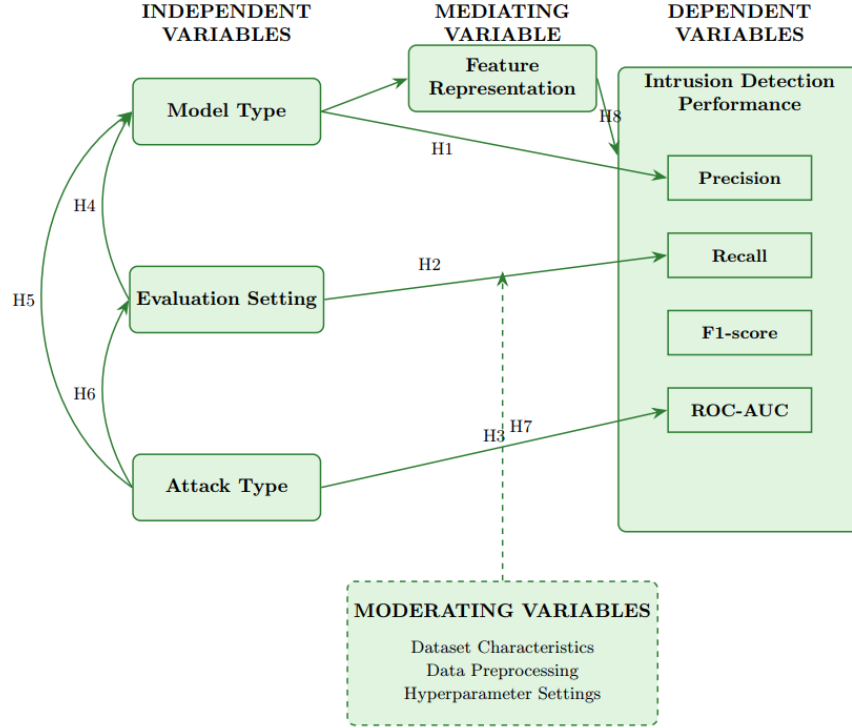


Figure 1: Conceptual Framework

2.8 Variables from Conceptual Framework

Independent Variables

Model Type: Supervised and Unsupervised Machine Learning Models for network intrusion detection. *Role:* Investigates the variations in performance based on model architecture.

Evaluation Setting: Evaluation environment (Within dataset and Cross-dataset). *Role:* Ascertains whether the environment has an impact cross dataset generalization capability of the model.

Attack Type: Specific types of attack contained in the dataset. *Role:* Investigates whether a specific type of attack favors detection performance.

Dependent Variables

Precision: The proportion of True Positives for attack detection. *Role:* Evaluates the degree of False Positives and the precision of positive attack predictions.

Recall: The proportion of actual attacks correctly detected by the model. *Role:* Evaluates the model's robustness to minimize missed attacks and identify attack instances.

F1_Score: The harmonic mean of Precision and Recall. *Role:* Provides a fair assessment of model performance by taking Precision and Recall into account.

ROC-AUC: Receiver Operating Characteristic - Area Under the Curve. *Role:* Evaluates how well a model can distinguish between various classes across categorization thresholds.

Moderating Variables

Dataset Characteristics: Characteristics of a dataset that includes data distribution, feature composition, class distribution, and attack representation. *Role:* Determines whether relationship between the independent variables and detection performance is affected by dataset differences.

Data Preprocessing: Data transformation - Cleaning, addressing missing values, feature encoding, scaling and other transformation techniques before training the models. *Role:* Moderates the relationship between individual variables and the performance output through modification and distribution of the data to the models.

Hyperparameter Setting: Tuning of hyperparameters for all the models. *Role:* Influences the learning process for the effectiveness of the model.

Mediating Variables

Feature Representation: Network traffic features supplied as inputs to the models are represented, transformed and arranged. *Role:* Explains how the link between model type and detection performance may be influenced by input feature qualities.

2.9 Research Objectives

The ultimate goal of this research is to examine the performances of supervised and unsupervised machine learning models, evaluate their capabilities in cross-dataset generalization for network intrusion detection. The research follows an experiment by training the models on CIC-IDS-2017 and evaluate their performance in CSE-CIC-IDS2018 and UNSW-NB15 datasets to identify different types of network attacks. We have set the following objectives to achieve our goal.

1. Process and standardize the CIC-IDS-2017, CSE-CIC-IDS2018 and UNSW-NB15 datasets.
2. To create a consistent feature-alignment procedure for CICIDS2017, CSE-CIC-IDS2018 and UNSW-NB15.
3. To develop a source-fitted preprocessing pipeline that prevent target-domain information leakage.
4. Train supervised and unsupervised machine learning models.
5. To use a held-out source validation partition for hyperparameter selection and anomaly-threshold calibration.
6. Evaluate the model performance on CSE-CIC-IDS2018 and UNSW-NB15 test datasets.
7. Make comparison of the evaluation metrics: F1_Score, MCC, ROC-AUC, Precision, and Recall.
8. To determine which model demonstrate the greatest robustness under dataset-distribution shift for intrusion detection in real-world cybersecurity environments.
9. To recommend improvement for reliable unsupervised cross-dataset intrusion detection.
10. Identify and select the best model that is more robust to dataset shift and generalizes better

Chapter 3

Methodology

We have implemented a quantitative approach for the study of Cross-Dataset Generalization for Network Intrusion Detection Using Supervised and Unsupervised Machine Learning Models. The study focuses on the overall performance of the models in detection of 'Attack' from the test datasets by training the models on an entire train dataset. Since the focus of this study is to investigate the cross-dataset generalization capabilities of both supervised and unsupervised machine learning models for network intrusion detection, the models will be trained entirely on CIC-IDS-2017 dataset and evaluated on unseen CSE-CIC-IDS2018 and UNSW-NB15 datasets.

The concept recognises that datasets are active components of the experiment rather than neutral containers. Existing reviews identify considerable variation in the realism, provenance, attack coverage, class balance, and feature construction of public NIDS datasets [Ring et al., 2019, Thakkar and Lohiya, 2020, Kenyon et al., 2020, Goldschmidt and Chudá, 2025, Pinto et al., 2025]. Harmonising feature meaning is necessary for cross-dataset transfer, but it does not make independently generated distributions identical. Corresponding features may retain different ranges, frequencies, and statistical relationships. These remaining differences form part of the generalisation challenge. Transfer direction must be retained explicitly because source and target distributions are not interchangeable. A model trained using CICIDS2017 and tested using UNSW-NB15 may behave differently from a model trained using UNSW-NB15 and tested using CICIDS2017.

This asymmetry occurs because each source dataset produces a different representation of normal traffic, preprocessing transformation, and decision threshold. Both directions must therefore be evaluated for each pair of datasets. Federated unsupervised detection provides a possible extensions to the conceptual framework. Learning from multiple decentralized sources could expose detector to more diverse normal behaviour without requiring organizations to centralise sensitive network traffic.

[Gourceyraud et al., 2025] combined federated learning with unsupervised clustering. This work connects unsupervised learning, privacy, and multi-source data. Federated learning is not evaluated in this dissertation, but it remains relevant to future privacy-preserving approaches for increasing the diversity of training data.

- **Research Type:** Quantitative
- **Data:** Publicly available datasets
 - **Train Dataset:** CIC-IDS-2017
 - **Test Dataset:** CSE-CIC-IDS2018 and UNSW-NB15
- **Sampling:** Non-Random Sampling
 - Convenience Sampling for Supervised Machine Learning Models
 - Purposive Sampling for Unsupervised Machine Learning Models
- **Programming Language:** Python 3.11
- **Libraries:** Pandas, Numpy, Tensorflow, Keras, Scikit-learn, Optuna, RobustScaler, Matplotlib, Seaborn, Pytorch
- **Models:** Random Forest, XGBoost, DNN, Isolation Forest, One-Class SVM, Improved Autoencoder
- **Editor:** MacTeX and TeXShop for preparing the dissertation

- **Evaluation Metrics:** F1-Score, MCC, ROC-AUC Curve

3.1 Cross-Dataset Generalization using Supervised Learning Models

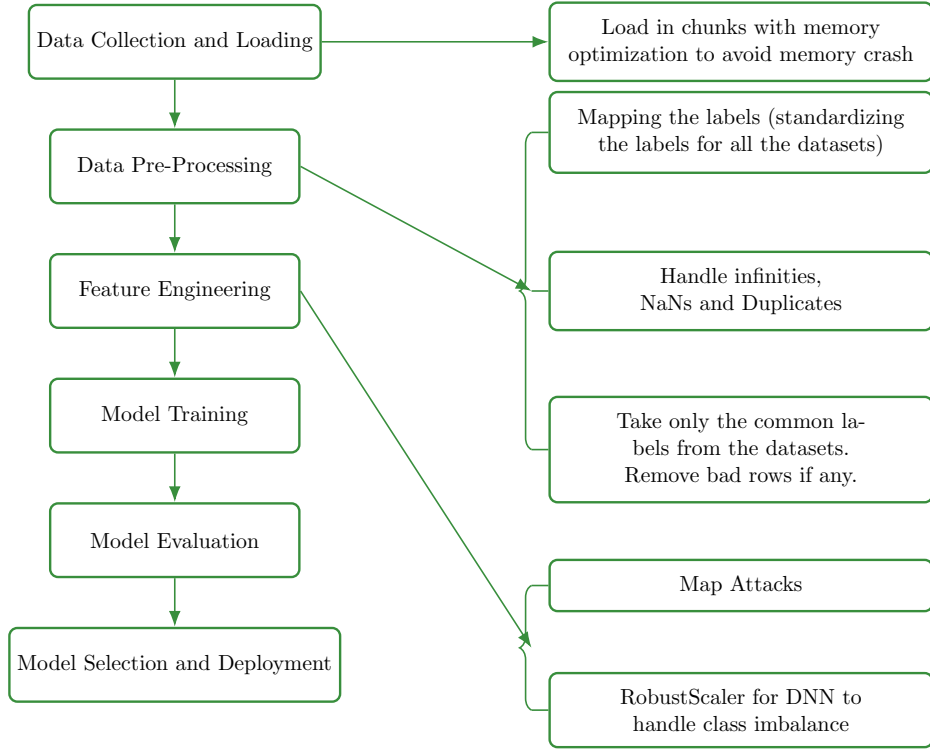


Figure 2: Workflow for Cross-Dataset Generalization for Network Intrusion Detection using Supervised Machine Learning Model.

3.1.1 Dataset Collection and Loading

For the study of Cross-Dataset Generalization for Network Intrusion Detection Using Supervised and Unsupervised Machine Learning Models at an academic level, we have sourced three datasets from public domain. The collected data are then loaded in chunks with memory optimization (dowcasting some datatypes) to avoid memory crash.

Table 2: Dataset Details

Dataset	Source	Link for the Source	Dataset Size	Dataset Shape	Access Date
CIC-IDS-2017	Kaggle	https://www.kaggle.com/datasets/chethuhn/network-intrusion-dataset	884.65 MB	2,830,743, 79	22/06/2026
CSE-CIC-IDS2018	Kaggle	https://www.kaggle.com/datasets/solarmainframe/ids-intrusion-csv	6.89 GB	16,233,002, 84	22/06/2026
UNSW-NB15	Canadian Institute for Cybersecurity	https://cicresearch.ca/CIC-Dataset/CIC-UNSW/	1.78 GB	3,540,241, 84	22/06/2026

CIC-IDS-2017 and CSE- CIC-IDS2018 datasets were downloaded from Kaggle. Meanwhile, the UNSW-NB15 dataset was fetched from Canadian Institute for Cybersecurity portal by submitting necessary information. We will employ CIC-IDS-2017 as the train dataset, while

CSE- CIC-IDS2018 and UNSW-NB15 will be our test datasets for cross-dataset generalization for NIDS. A quick exploration of these datasets before data pre-processing showed dominant presence of ‘Benign’ class in all the datasets. 2273097 (80.3%) in CIC-IDS-2017, 13484708 (83.06%) in CSE- CIC-IDS2018 and 3450658 (97.46%) in UNSW-NB15 are labeled as ‘Benign’ indicating huge class imbalance between the benign and attack classes.

CIC-IDS-2017

The dataset consists of network traffics collected from Monday to Friday at different time intervals originally extracted by the Canadian Institute for Cybersecurity for the study of network intrusion detection. CIC-IDS-2017 datasets were generated utilizing CICFlowMeter to create network flows and labelled them according to timestamps, source and target IP address, source and target ports, communication protocol and attack types. To prepare a realistic background traffic, Canadian Institute for Cybersecurity (CIC) applied the suggested B-Profile system [Sharafaldin et al., 2018] to model the abstract patterns of human interactions and generate the realistic benign background traffic from 25 real users. The 8 files combined has 79 features and 2830743 rows.

CSE-CIC-IDS2018

Among all the datasets used for our research, this one is the biggest in size (10 files of 6.89 GB) with more than 16 million rows and 84 feature downloaded through Kaggle. CSE-CIC-IDS2018 was a collaborative initiative between CIC and Communications Security Establishment (CSE) with the objective of creating standard benchmark datasets for intrusion detection systems to properly detect network anomalies. The dataset was generated based on different attack types such as Botnets, Brute force, DoS and Heartbleed by accessing different user profiles. The dataset is comprised of 80 features obtained through incoming network traffic and system logs with CICFlowMeter-V3 tool with proper utilization of B-profile and M-profile to identify the statistical user behavior through the reproduction of network traffics and attack scenarios. The initiative used Slowloris and LOIC to inject HTTP DoS attacks, HOIC for DDoS, and bombarded web applications with Damn Vulnerability Web App (DVWA) to test the vulnerabilities.

UNSW-NB15

CIC-UNSW-NB15 or simply UNSW-NB15 augmented dataset was created with the help of IXIA PerfectStorm with up-to-date normal and abnormal network traffic. The dataset contains 9 types of attacks which the CIC-IDS-2017 and CSE- CIC-IDS2018 datasets lack. CICFlowmeter extracted new features from the network flow. The attacks in UNSW-NB15 are categorized by matching the flow records: source/destination IP, port, and protocols with the records from ground truth file. UNSW-NB15 prioritized matching Timestamp in-case of multiple records, removed flows that could not be categorized and categorized the remaining traffic as benign after malicious flows were processed.

3.1.2 Data Pre-Processing

The first step in data-processing is standardizing all the labels across the datasets. We mapped the labels of the datasets taking reference to CSE-CIC-IDS2018. We handled the infinite values in these datasets by replacing them with NaN values. Removing duplicates earlier would significantly reduce the size of the datasets, seeing the skewness of the datasets, we replaced the NaN values with the median values from the respective labels. And, only after replacing the NaN values with median values, we removed the duplicated values from the datasets. Finally, for the next step we took all the common labels from the datasets. We also checked and deleted the bad rows from the datasets.

Table 3: Attack Class Distribution before Data Pre-Processing

Label	CICIDS2017	CSE-CIC-IDS2018	UNSW-NB15
benign	2273097	13484708	3450658
bot	1966	286191	0
ddos	128027	0	0
dos goldeneye	10293	0	0
dos hulk	231073	0	0
dos slowhttpptest	5499	0	0
dos slowloris	5796	0	0
ftp-patator	7938	0	0
heartbleed	11	0	0
infiltration	36	0	0
portscan	158930	0	0
ssh-patator	5897	0	0
web attack – brute force	1507	0	0
web attack – sql injection	21	0	0
web attack – xss	652	0	0
brute force -web	0	611	0
brute force -xss	0	230	0
ddos attack-hoic	0	686012	0
ddos attack-loic-udp	0	1730	0
ddos attacks-loic-http	0	576191	0
dos attacks-goldeneye	0	41508	0
dos attacks-hulk	0	461912	0
dos attacks-slowhttpptest	0	139890	0
dos attacks-slowloris	0	10990	0
ftp-bruteforce	0	193360	0
infiltration	0	161934	0
label	0	59	0
sql injection	0	87	0
ssh-bruteforce	0	187589	0
analysis	0	0	385
backdoor	0	0	452
dos	0	0	4467
exploits	0	0	30951
fuzzers	0	0	29613
generic	0	0	4632
reconnaissance	0	0	16735
shellcode	0	0	2102
worms	0	0	246

Table 4: Attack Class Distribution after Data Pre-Processing

Label	CICIDS2017	CSE-CIC-IDS2018	UNSW-NB15
benign	2073870	12196127	3389125
bot	1953	144535	0
ddos	128016	0	0
dos goldeneye	10286	0	0
dos hulk	172849	0	0
dos slowhttpptest	5228	0	0
dos slowloris	5385	0	0
ftp-patator	5933	0	0
heartbleed	11	0	0
infiltration	36	0	0
portscan	90819	0	0
ssh-patator	3219	0	0
web attack – brute force	1470	0	0
web attack – sql injection	21	0	0
web attack – xss	652	0	0
brute force -web	0	553	0
brute force -xss	0	228	0
ddos attack-hoic	0	198861	0
ddos attack-loic-udp	0	1730	0
ddos attacks-loic-http	0	576191	0
dos attacks-goldeneye	0	41406	0
dos attacks-hulk	0	145199	0
dos attacks-slowhttpptest	0	55	0
dos attacks-slowloris	0	9908	0
ftp-bruteforce	0	54	0
infiltration	0	140612	0
label	0	1	0
sql injection	0	84	0
ssh-bruteforce	0	94048	0
analysis	0	0	385
backdoor	0	0	452
dos	0	0	4459
exploits	0	0	30951
fuzzers	0	0	29598
generic	0	0	4605
reconnaissance	0	0	16710
shellcode	0	0	2102
worms	0	0	246

3.1.3 Feature Engineering

In order to maximize the intrusion detection, we mapped the attacks of similar nature. Since, the tree based models are robust in handling the class-imbalance, we have simply implemented class-weighting and avoided using SMOTE for Random Forest and XGBoost. Meanwhile, for our DNN architecture, we used RobustScaler to fit-transform the train datasets only. To check the best hyperparameters for the models, we have utilized Optuna study [Akiba et al., 2019] to get the best configuration of the models that would generate maximum F1 score for the 'Attack' class .

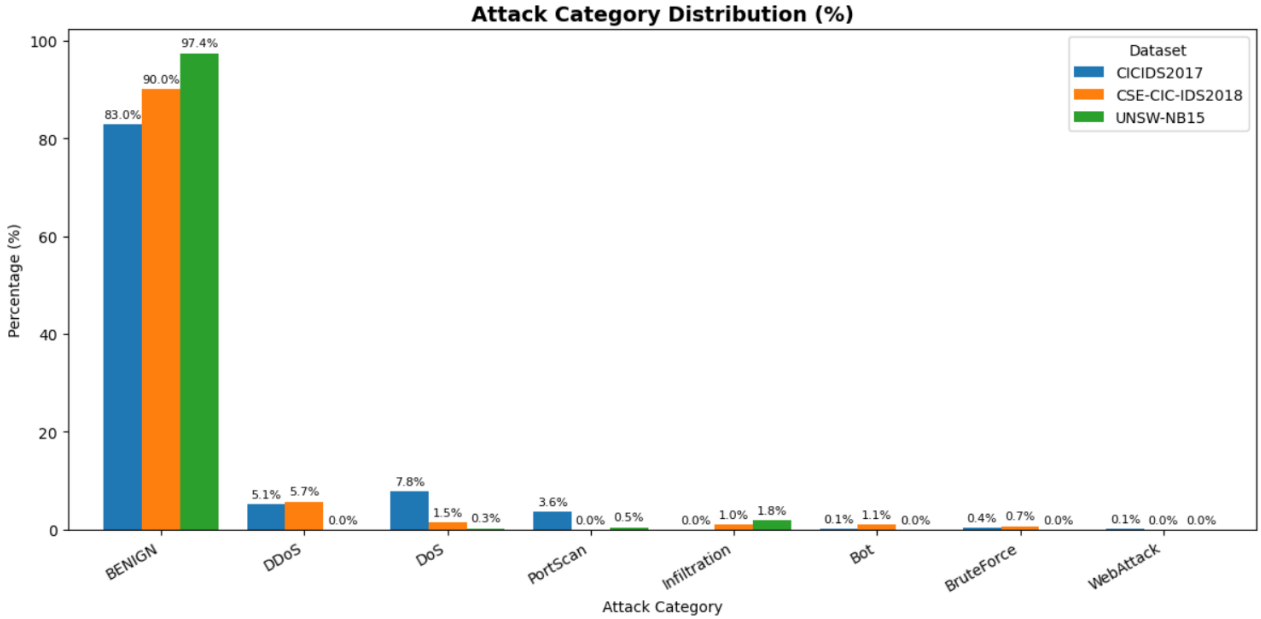


Figure 3: Attack Distribution across Datasets after Mapping 'Attack'

Table 5: Attack Counts across Datasets after Mapping 'Attack'

Label	CICIDS2017	CSE-CIC-IDS2018	UNSW-NB15
BENIGN	2073870	12196127	3389125
Bot	1953	144535	0
BruteForce	9152	94655	0
DDoS	128016	776782	0
DoS	193748	196568	9064
Infiltration	47	140612	63734
PortScan	90819	0	16710
WebAttack	2143	312	0

3.1.4 Model Training

Random Forest

Random Forest is a robust ensemble learning model that creates many decision trees and carries out classification by taking the mode of the predictions of individual trees. During the training phase, only a random subset of features is considered at each node split. The trees in a random forest use a bootstrap-sampled version of the training data to introduce randomness in both the sampling process and feature selection, thereby minimizing correlation among the trees and reducing the variance of the ensemble without increasing bias.

The significance of tree-based models lies in their ability to classify the feature space using threshold splits without being affected by monotonic transformations of the input features. In this study, we trained Random Forest on the full CIC-IDS-2017 dataset without scaling, despite the substantial class imbalance. As Random Forest is resistant to overfitting on high-dimensional flow-related data and shows robustness to noisy features, the workflow avoids Synthetic Minority Over-Sampling Technique (SMOTE) and instead uses class weight = "balanced" to handle class imbalance.

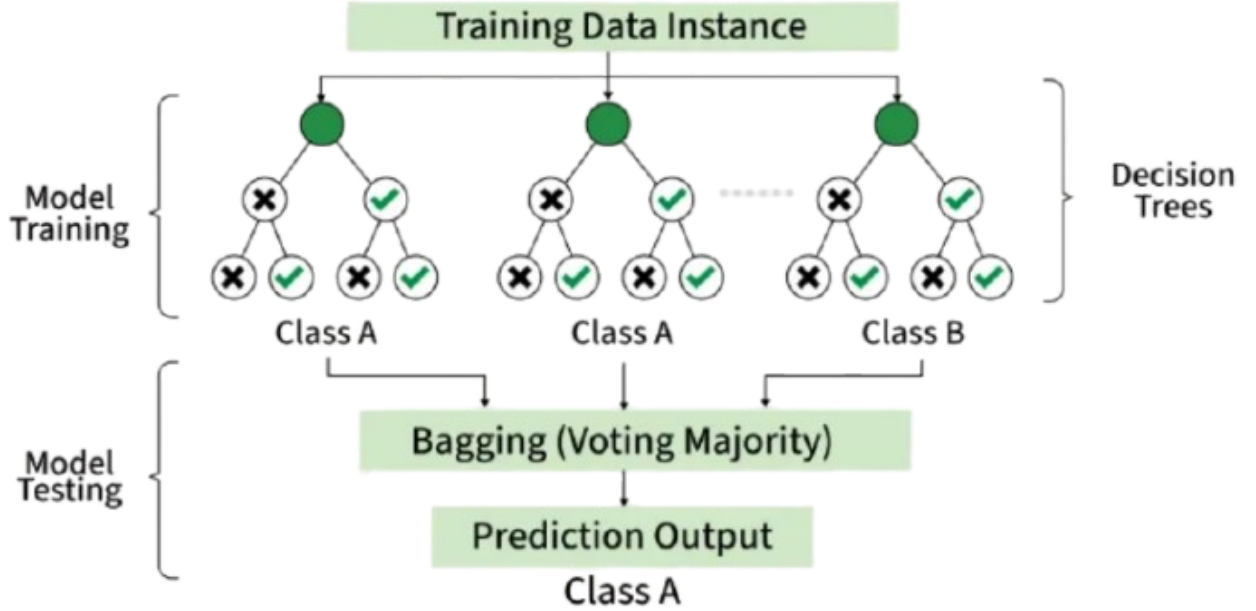


Figure 4: Random Forest Architecture

Table 6: Hyperparameters for Random Forest

Hyperparameter	Value	Purpose
n_estimators	200	Number of trees in the forest
max_depth	10	Maximum depth of each tree
min_samples_split	2	Minimum samples to split an internal node
min_samples_leaf	2	Minimum samples for each leaf node
max_features	sqrt	Number of features at each split
bootstrap	True	Bootstrap sampling to build trees
criterion	gini	Function to measure split quality
class_weight	balanced	To automatically address the class imbalance
random_state	42	Seed for reproducibility
n_jobs	-1	Parallel Processing

XGBoost

eXtreme Gradient Boosting generally termed as XGBoost is a scalable gradient boosting decision tree which boost weak learners. XGBoost utilizes the technique where each tree is trained to rectify the mistakes of previous trees. It does not build trees based on parallel bagging as Random Forest does. Rather XGBoost allows trees to grow in additive way to optimize the differential loss function using gradient descent. It employs L1 and L2 regularization to control overfitting and complexity. XGBoost inhibits computational effectiveness and robustness in solving structured data and is capable to construct parallel tree to deal with the missing data.

So, with its exceptional ensemble technique to compare methods of ensemble trees that differ in aggregation approach (sequential vs parallel), we have selected XGBoost along with Random Forest as our benchmark supervised learning models. Like random forest, we have implemented class-weighting to mitigate the class imbalance between the benign and malicious cases in the training set, i.e. CIC-IDS-2017.

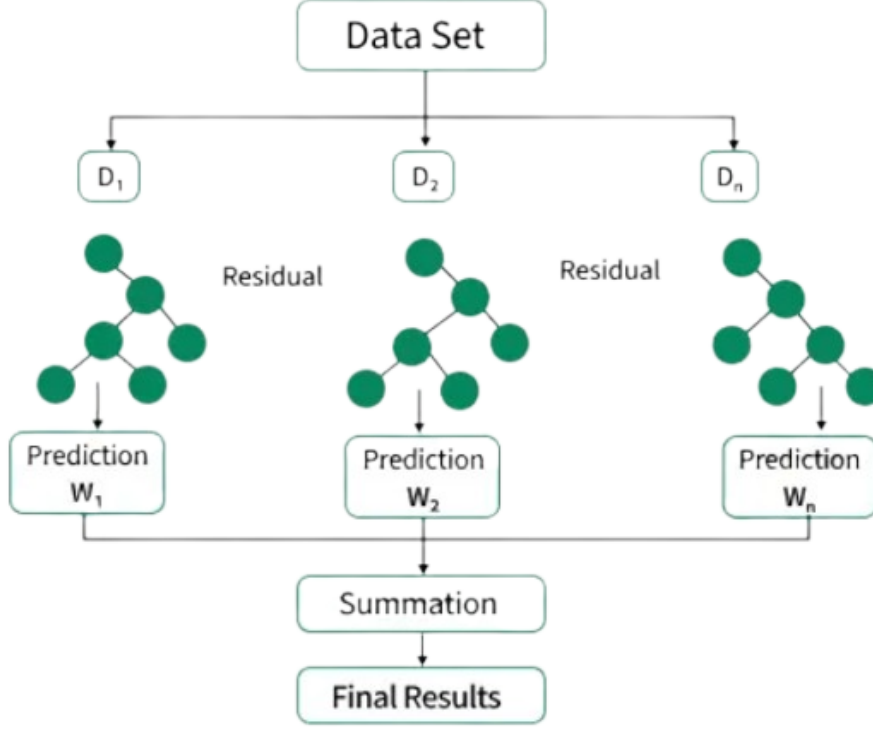


Figure 5: XGBoost Architecture

Table 7: Hyperparameters for XGBoost

Hyperparameter	Value	Purpose
n_estimators	200	Number of boosting trees in the forest
max_depth	6	Maximum depth of each tree
learning_rate	0.1	Contribution of each tree to the final model
sub_sample	0.8	Training sub-samples to boost iteration
col_sample_bytree	0.8	Fraction of features for each tree
objective	binary:logistic	Specifies binary classification
eval_metric	logloss	Metric to calculate classification error
tree_method	hist	Histogram based training
scale_pos_weight	neg/pos	Handles class imbalance
random_state	42	Seed for reproducibility
n_jobs	-1	Parallel Processing

Deep Neural Network (DNN)

DNN learn feature interactions without manual feature extraction giving it an extra edge over tree-based ensemble techniques. DNN is dependent on the size/amount of data, thus shows better performance as the degree of changes in inputs rises. DNN uses hidden layers for classification. Each hidden layer utilizes ReLU activation to bring in non-linearity after each hidden layer to prevent the vanishing gradient problem of deeper models.

To avoid memory crash given the size of the dataset, we tested on 20% of the CIC-IDS-2017 with tuning size of 300000. Considering the computational cost and to prevent memory crash, we have assigned `n_trials` to 30 only. We have implemented 4 hidden layers with 512, 256, 128 and 64 neurons for binary classification “benign” and “attack” in this study. We have implemented RobustScaler for scaling the datasets to mitigate the effect of strong outliers (flow duration, packet inter-arrival times). Similar to Random Forest and XGBoost which utilizes class-weight, weighted loss in DNN handles class imbalance. In consideration to the huge test dataset size, prediction and evaluation of the model has been performed in batches of max 4096. The `best_dnn_params` identifies the best configuration of the model specifically the number of layers, units per layer, dropout rate and the learning rate to construct the best DNN architecture. Thus, the evaluation using DNN architecture works in two phases, first finding the best configuration by training and testing the model on CIC-IDS-2017 and secondly using the best hyperparameters to evaluate the performance of the model on the test datasets CSE-CIC-IDS2018 and UNSW-NB15.

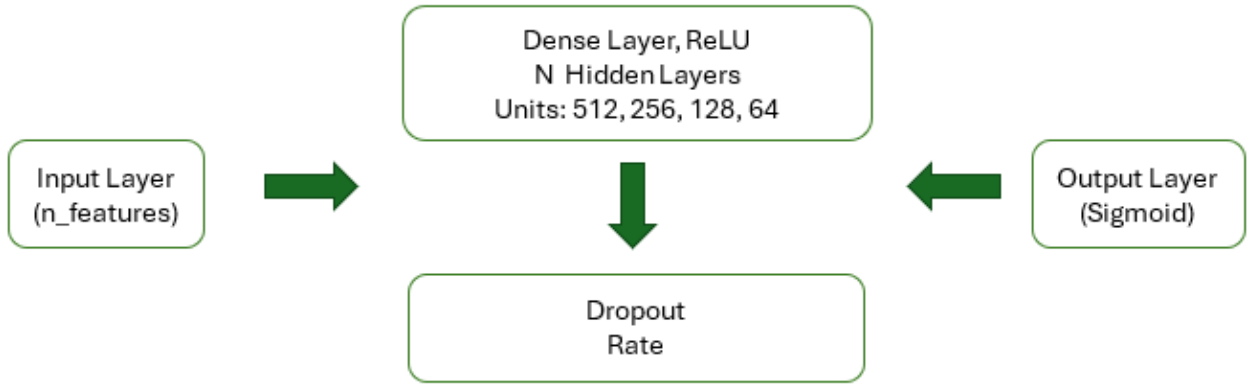


Figure 6: DNN Architecture

3.2 Cross-Dataset Generalization using Unsupervised Machine Learning Models

The following workflow proposes that anomaly-detection performance is influenced by the selected model, evaluation setting, source dataset, target dataset, and direction of transfer. Dataset shift moderates these relationships because differences between source and target traffic alter the distributions of the features used by the detector.

Isolation Forest, One-Class SVM, and the improved autoencoder represent normal traffic differently. Isolation Forest measures how easily observations can be isolated. One-Class SVM estimates a boundary around normal traffic. The improved autoencoder measures reconstruction error. Evaluation setting plays significant role in the performance of models for intrusion detection. Within-dataset evaluation tests a model using held-out observations originating from the same dataset as the training data. Cross-dataset evaluation applies a source-fitted model to independently collected target traffic.

The relationships are controlled through a harmonised features space, source-only preprocessing, consistent data partitions, source-validation calibration, common metric definitions, and the prohibition of target-domain training or threshold calibration.

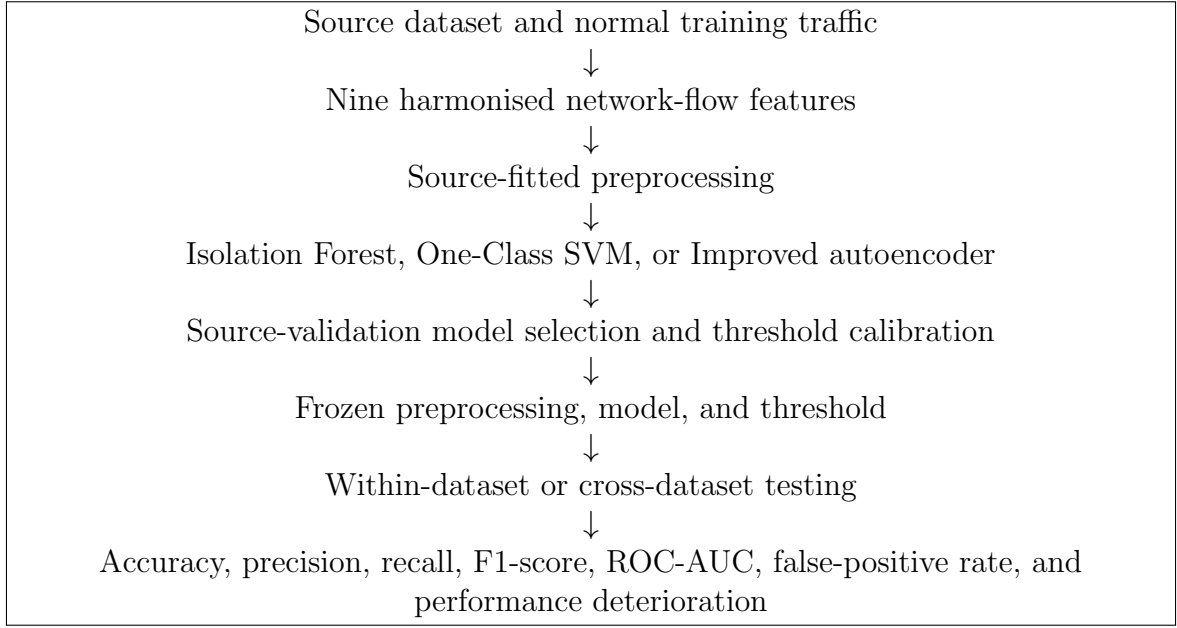


Figure 7: Workflow for Cross-Dataset Generalization for Network Intrusion Detection using Unsupervised Machine Learning Model.

The workflow integrates one-class learning, three alternative representation of normal traffic, and cross-domain distribution shift.

For a source dataset D_s , a detector learns an anomaly-scoring functions $f_s(x)$. A source - validations procedures selects an operating threshold τ_s . The fitted preprocessing procedure, anomaly-scoring functions, and threshold are then frozen and applied unchanged to a target dataset D_t .

Cross-dataset robustness is reflected in the extent to which the source-derived anomaly scores continues to distinguish benign traffic from attack traffic in D_t .

Purposive sampling was used to select the three datasets. They were selected because they are established intrusion-detection benchmarks, contain labelled benign and attack traffic, and provide sufficient flow information to construct a common semantic feature space.

The inclusion of CICIDS2017 and CSE-CIC-IDS2018 enabled the study to evaluate transfer between two dataset produced by related institutions. The inclusions of UNSW-NB15 introduced a benchmark created using a different network environment and feature schema.

This combination enabled the investigation to compare transfer between related dataset with transfer involving a more structurally different dataset. [Ring et al., 2019, Thakkar and Lohiya, 2020, Kenyon et al., 2020],

3.2.1 Stratified Data Partitioning

Each dataset was divided using stratified random partitioning so that the proportions of benign and attack observations were approximately preserved.

First, 20% of each dataset was reserved as the final test partitions. The remaining 80% formed the development partition

Second, 20% of the development partition was reserved for validation. Before sample caps were applied, the effective allocation was therefore approximately;

- 64% training pool
- 16% validation partition
- 20% final test partition

Only benign observation from the training pool was retained for model fitting. Both benign and attack observation were retained in the source-validation and final test partitions.

The validation partition was used for source -domain model selection and threshold calibration. The partition remained independent of model development and was used only for final evaluation.

3.2.2 Bounded Sampling

Computationally bounded samples were used to maintain feasible and consistent experiments. The following maximum sample sizes were defined:

- normal training data: 100,000 observations
- validation data: 50,000 observations
- final test data: 80,000 observation
- One-Class SVM training data: 8,000 normal observation

When a validation or test partition exceeded its corresponding cap, stratified sampling was used to retain approximately the same class distribution.

Normal-Training and One-Class SVM samples were selected without replacement using seeded random sampling. These caps were introduced as computational controls and did not permit observations to be shared across training, validation, and testing partitions.

3.2.3 Autoencoder Validation Sample

Normal observations from the source-validation partition were identified and shuffled using seeded random generator. A held out portion of those normal observation was used to monitor reconstruction loss and apply early stopping.

The complete labelled source-validation partition was used separately to calculate anomaly scores and select the final decision threshold.

3.2.4 Random-State Procedure

A base random state of 42 was used. Deterministic dataset-specific offsets were applied during partitioning and subsampling so that the same procedure could be reproduced could be reproduced while avoiding identical random sequences for every dataset.

The random state controlled ;

- stratified development and test partitioning
- stratified training and validation partitioning
- normal-training subsampling
- validation and test sample capping
- One-Class SVM training-sample selection

- Isolation forest construction
- auto encoder parameter initialisation.

3.2.5 Model Training

Isolation Forest

Isolation Forest is a tree-based anomaly detectors that recursively partitions the features space using randomly selected features and split values. Rare or structurally unusual observations tend to be isolated using shorter tree paths than observations located in dense regions.

An ensemble of isolation trees converts the average path length into an anomaly score. Shorter average paths indicates a greater degree of anomaly.

The method is suitable for tubular network-flow data because it is scalable and does not require pairwise distance calculations between all observations. Random subsampling also reduces its computational requirements.

$$s(x, n) = 2^{-\frac{E[h(x)]}{c(n)}}, \quad (3.1)$$

where $E[h(x)]$ is the expected path length of x across the isolation trees and $c(n)$ is the normalizing path length for a sample containing n observations. Values closer to one indicate stronger anomalies.

However, the method’s definitions of isolation depends on the distribution learned from the source data. Legitimate target observation may be isolated quickly when their feature values fall outside the ranges observed in the source dataset. Isolation Forest may therefore achieve high recall while generating a large number of false positives under severe distribution shift. Accuracy alone cannot reveal this trade-off. Precision, recall, F1_Score, and false-positive rate must therefore be interpreted together [Verkerken et al., 2022, Yang et al., 2022].

Isolation Forest implementation:

- 350 isolation trees;
- automatic contamination estimation;
- parallel processing;
- a fixed base random state of 42; and
- three candidate maximum-sample values.

The candidate maximum-sample values were:

$$m \in \{512, 2048, 8192\}. \quad (3.2)$$

When a candidate exceeded the available number of normal-training observations, the complete available sample was used.

Each candidate was fitted using normal source traffic. Its anomaly scores were calculated for the labelled source-validation partition. The candidate producing highest source-validation F1-score was selected.

The decision function was inverted so that larger scores represented stronger anomalies:

$$s_i^{IF} = -f_{\text{decision}}(x_i). \quad (3.3)$$

The selected Isolation Forest model and its source-calibrated threshold were frozen within-dataset and cross-dataset testing.

One-Class Support Vector Machine

One-Class SVM is a boundary-based anomaly detector. It maps normal training observations into a transformed features space using a kernel functions and estimates a boundary containing most of those observations.

The model attempts to separate the transformed normal observations from the origin with the greatest possible margin. A simplified objective functions is expressed as:

$$\min_{w, \rho, \xi} \left[\frac{1}{2} \|w\|^2 + \frac{1}{\nu n} \sum_{i=1}^n \xi_i - \rho \right], \quad (3.4)$$

subject to:

$$w^T \phi(x_i) \geq \rho - \xi_i, \quad \xi_i \geq 0, \quad (3.5)$$

where $\phi(x_i)$ maps observation x_i into the kernel feature space, ρ determine the decision boundary, ξ_i represents a slack variable, and ν controls the permitted proportion of training observation outside the estimated boundary.

The radial basis functions kernel allows the model to represent a non-linear normal-traffic boundary. It is expressed as:

$$K(x_i, x_j) = \exp \left(-\gamma \|x_i - x_j\|^2 \right), \quad (3.6)$$

where γ controls the influence of individual training observations.

The model's behavior is influenced by feature scaling, γ , ν , and the size of the training sample. One-Class SVM is therefore sensitive to preprocessing decisions and hyperparameter selection.

A boundary fitted tightly around the source distribution may reject large amount of legitimate target traffic. Conversely, an overly broad boundary may accept malicious observations. Its computational and memory requirements may also increase substantially as the training sample grows [Verkerken et al., 2022, Khraisat et al., 2019].

One Class SVM was fitted using a maximum of 8,000 randomly selected normal source observations. Bounding the training sample reduced the computational and memory requirements of kernel model.

A radial basis function kernel was employed. Two candidate configurations were evaluated:

- $\nu = 0.03$ with $\gamma = \text{scale}$; and
- $\nu = 0.08$ with $\gamma = 0.05$.

Each candidate was fitted using the same bounded normal source sample and evaluated using the source-validation partition. The configuration producing the highest validation F1_Score

was selected. The decision function was inverted so that the larger values represented stronger anomalies.

$$s_i^{\text{SVM}} = -f_{\text{decision}}(x_i). \quad (3.7)$$

The selected model and source-calibrated threshold were applied unchanged to every target dataset associated with the source domain.

Improved Autoencoder

An autoencoder consists of an encoder and a decoder. The encoder maps an input observations to lower-dimensional latent representation:

$$z_i = f_{\theta}(x_i), \quad (3.8)$$

where x_i is the input observation, z_i is its latent representation, and θ represents the encoder parameters.

The decoder attempts to reconstruct the original input:

$$\hat{x}_i = g_{\phi}(z_i), \quad (3.9)$$

where ϕ represents the decoder parameters. When the autoencoder is fitted using normal traffic, it is expected to reconstruct observation resembling the training data more accurately than anomalous observations.

For an observations x_i and its reconstruction $\hat{x}_i$, the mean-squared reconstruction error is:

$$s_i^{AE} = \frac{1}{d} \sum_{j=1}^d (x_{ij} - \hat{x}_{ij})^2, \quad (3.10)$$

where d is the number of input features. A higher reconstruction error indicates a greater departure from the learned representation of normal traffic.

Deep representation learning support the modeling of non-linear relationships that may be difficult for shallow methods to represent. However, additional model capacity does not automatically improve anomaly detection. A highly expressive decoder may reconstruct attack observation accurately, while an overfitted latent representation may encode pattern specific to the source dataset. Regularization architecture selection, validation and early stopping are therefore important [Pang et al., 2021, Yang et al., 2022]. The improved autoencoder used in the present study employs a deeper regularized architecture and early stopping to reduce overfitting.

These improvement do not eliminate distribution shift. Normal target traffic may still produce large reconstruction error when its patterns differ substantially from the source-domain data. The improved auto encoder was implemented using PyTorch. For input dimension d , the hidden-layer size was defined as:

$$h = \max(16, 2d), \quad (3.11)$$

and the bottleneck dimension was defined as :

$$b = \max \left(4, \left\lfloor \frac{d}{2} \right\rfloor \right). \quad (3.12)$$

because the experiment used nine input features, the resulting architecture was:

$$9 \rightarrow 18 \rightarrow 4 \rightarrow 18 \rightarrow 9. \quad (3.13)$$

The encoder applied a linear layer, batch normalization, Leaky ReLU activation, five-per-cent dropout, and a second linear layer leading to the bottleneck. The decoder reconstructed the input through a hidden layer and output layer.

The model was optimized using AdamW with:

- learning rate 8×10^{-4} ;
- weight decay 1×10^{-5} ;
- training batch size of 1,024;
- maximum of 25 epochs; and
- early-stopping patience of six epochs.

Smooth L1 loss was used for training because it is less sensitive to extreme reconstruction errors than ordinary mean-squared loss. Early stopping monitored reconstruction loss using held out normal source-validation observations. The model parameters producing the lowest validation loss were restored after training.

The final anomaly score was the mean-squared reconstruction error:

$$s_i^{AE} = \frac{1}{d} \sum_{j=1}^d (x_{ij} - \hat{x}_{ij})^2. \quad (3.14)$$

A higher reconstruction error indicated a greater departure from the normal source-domain representation.

3.2.6 Threshold Calibration

Each model produced a continuous anomaly score. A threshold was required to convert that score into a binary prediction:

$$\hat{y}_i = \begin{cases} 1, & \text{if } s_i \geq \tau_s, \\ 0, & \text{if } s_i < \tau_s. \end{cases} \quad (3.15)$$

The threshold producing the highest source-validation F1_Score was selected:

$$\tau_s = \arg \max_{\tau} F1(y_s^{\text{val}}, \mathbb{I}[s_s^{\text{val}} \geq \tau]). \quad (3.16)$$

The selected source threshold was frozen before testing. Target labels were not used to select, modify, or recalibrate it.

F1_Score measured the quality of binary predictions produces by the source calibrated threshold. ROC-AUC measured the ability of the continuous scores to rank attack observations above benign observations across possible thresholds.

Performance deterioration was calculated by comparing a cross-dataset result with the corresponding source model's within-dataset baseline:

$$\text{Deterioration} = \frac{M_{\text{within}} - M_{\text{cross}}}{M_{\text{within}}} \times 100, \quad (3.17)$$

where M represents F1_Score or ROC-AUC. A positive value indicated performance deterioration during cross-dataset transfer, whereas a negative value indicated that cross-dataset performance exceeded the corresponding within-dataset baseline.

3.2.7 Model Evaluation

Since, the main objective of the study is to find out which model shows robust performance in cross-dataset generalization for network intrusion detection, we have prioritized F1-Score, MCC and ROC-AUC Curve as the evaluation metrics. Accuracy alone cannot define whether the model performed better in detecting 'Attack' from the network traffic.

- Precision: It gives the measure of the proportion that are predicted as Positive. It is the ratio of True Positives (TP) to the summation of True Positives and False Positives (FP).

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Recall: Unlike Precision, it gives the measure of positives that are correctly predicted as Positive. It demonstrates the model's proficiency in detecting True Positives while limiting False Negatives (FN). It is the ratio of True Positives (TP) to the summation of True Positives and False Negatives (FN).

$$\text{Recall} = \frac{TP}{TP + FN}$$

- F1_Score: It is the harmonic mean of Precision and Recall giving a balanced measure that shows the overall performance of the model minimizing both False Positives and False Negatives.

$$\text{F1_Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- MCC: Matthews Correlation Coefficient or simply MCC is a metric used to evaluate the classification performance of the model by determining the correlation between actual and predicted classes, preferable metric for evaluating model performance on imbalanced datasets. Its value ranges from -1 to 1; 1 indicates strong prediction, 0 indicates random prediction and -1 represents weak prediction between actual and predicted classes.

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

- ROC-AUC: Receiver Operating Characteristic Area Under Curve (ROC-AUC) gives the measure specifying how well the model performed in distinguishing the positive and negative classes into correct labels with respect to various classification thresholds. The value ranges from 0 to 1; higher the value better the classification of the classes.

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR})$$

Chapter 4

Evaluation and Results

Table 8: Supervised Machine Learning Model Performance Metrics without Optuna Study

Model	Test Dataset	Accuracy	Precision	Recall	F1	Macro F1	ROC-AUC	MCC
Random Forest	CSE-CIC-IDS2018	0.9188	0.9239	0.2043	0.3346	0.6457	0.8384	0.4130
	UNSW-NB15	0.9743	0.2830	0.0002	0.0003	0.4936	0.7613	0.0063
XGBoost	CSE-CIC-IDS2018	0.9309	0.9799	0.3150	0.4768	0.7199	0.9153	0.5346
	UNSW-NB15	0.9752	0.6533	0.0739	0.1328	0.5601	0.7829	0.2142
Deep Neural Network	CSE-CIC-IDS2018	0.8932	0.1393	0.0133	0.0244	0.4839	0.5346	0.0129
	UNSW-NB15	0.8820	0.0492	0.1958	0.0787	0.5078	0.8107	0.0501

All the supervised models generated accuracy greater than 88%. Random Forest and XGBoost yielded Precision score above 90% for the CSE-CIC-IDS2018 dataset. XGBoost turned out to be the best performing model while evaluated on the same dataset with Recall Score of 0.31, F1-Score of 0.47, MCC of 0.53 compared to Recall Score of 0.20, F1-Score of 0.33 and MCC of 0.41 generated by Random Forest. These scores achieved are better than the scores achieved from previous studies 1. The performance of these models dropped significantly when evaluated on UNSW-NB15 dataset. The only significant metric being MCC which is 0.21 for XGBoost. The primary reason for the performance drop could be due to variation in the types of 'Attack' in the dataset 3. The 'Attack' labels that are present in the train dataset CIC-IDS-2017 are not present in the test dataset UNSW-NB15 and vice-versa. The study of [Chandra and Anandappa, 2025] have highlighted the models' robustness in cross-dataset generalization for network intrusion detection while testing on this particular dataset. Also, the methodology used are in contrast to what we have adopted.

The biggest disappointment above all is the result from DNN architecture. The model yielded accuracy above 88% for both datasets, however the results for our preferred metrics are negligible/avoidable. As, DNN relies on the depth of the layers to learn complex features and their patterns, huge class-imbalance and the depth of layers used could have been the reason for the poor performance. We have only used 4 hidden layers to detect attacks from the network traffic. Further more, we conducted Optuna study twice to tune the hyperparameters and used them to check whether the results improve, there was no significant change in the results of Random Forest and XGBoost, however we saw improvement in the performance of DNN. But tuning the hyperparameters showed inconsistent influence over the test datasets CSE-CIC-IDS2018 and UNSW-NB15.

The following are the hyperparameters we used for our Optuna Studies.

Table 9: Hyperparameters for DNN Optuna Studies

Hyperparameter	Search Space	Purpose
n_layers	2, 6	Controls network depth, learns complex features and balances overfitting
units per layer	[512, 256, 128, 64]	Controls width of hidden layer and checks the representational capacity at each layer
learning_rate	[1e-4, 1e-2]	Controls the step size during gradient descent
dropout_rate	0.1, 0.5	Deactivate the neurons during training to prevent overfitting and overall regularizes the network
batch_size	[512, 1024, 2048, 4096]	Prevents memory crashes, training speed and gradient noise by controlling the number of samples processed

Table 10: DNN Performance Metrics with Optuna Study -1

Model	Test Dataset	Accuracy	Precision	Recall	F1	Macro F1	ROC-AUC	MCC
Deep Neural Network	CSE-CIC-IDS2018	0.9001	0.4135	0.0005	0.0010	0.4742	0.5954	0.0114
	UNSW-NB15	0.9759	0.8357	0.0794	0.1449	0.5664	0.8096	0.2532

The best hyperparameters after Optuna Study 1 were:

$$\{n_layers, units, dropout_rate, learning_rate, batch_size\} = \{4, 512, 0.179447, 0.001116, 512\}.$$

Table 11: DNN Performance Metrics with Optuna Study -2

Model	Test Dataset	Accuracy	Precision	Recall	F1	Macro F1	ROC-AUC	MCC
Deep Neural Network	CSE-CIC-IDS2018	0.9058	0.9824	0.0584	0.1102	0.5302	0.5470	0.2274
	UNSW-NB15	0.9360	0.0025	0.0038	0.0030	0.4850	0.7923	-0.0293

The best hyperparameters after Optuna Study 2 were:

$$\{n_layers, units, dropout_rate, learning_rate, batch_size\} = \{4, 512, 0.111666, 0.000527, 512\}.$$

Table 12: Unsupervised Machine Learning Model Performance Metrics

Model	Test Dataset	Accuracy	Precision	Recall	F1	ROC-AUC
Isolation Forest	CSE-CIC-IDS2018	0.7409	0.3138	0.4469	0.3687	0.6947
	UNSW-NB15	0.3019	0.1344	0.8306	0.2314	0.3528
One-Class SVM	CSE-CIC-IDS2018	0.7851	0.1585	0.0625	0.0897	0.7241
	UNSW-NB15	0.5948	0.0194	0.0444	0.0270	0.4193
Improved Auto-Encoder	CSE-CIC-IDS2018	0.7129	0.2237	0.2818	0.2494	0.6534
	UNSW-NB15	0.5063	0.1844	0.8481	0.3029	0.7573

Unlike the performance of supervised machine learning models, all the three unsupervised machine learning models generated accuracy lesser than 80% for CSE-CIC-IDS2018 dataset and below 60% while tested on UNSW-NB15 dataset. Isolation Forest showed superior performance while generalizing for CSE-CIC-IDS2018 dataset with F1_Score of 0.3687 and ROC-AUC score of 0.6947. Meanwhile, Improved Auto-Encoder outperformed other three unsupervised models while evaluating the UNSW-NB15 dataset with F1_Score of 0.3029 and ROC-AUC score

of 0.7573. Unlike XGBoost, which turned out to be the most robust model for cross-dataset generalization (considering both test datasets), no single unsupervised machine learning models showed superior capability for cross-dataset generalization for network intrusion detection. One-Class SVM turned out to be the least performing unsupervised model for cross dataset generalization with F1_Score of 0.0897 and 0.0270 for CSE-CIC-IDS2018 and UNSW-NB15 datasets respectively. Furthermore, the results of both the supervised and unsupervised machine learning models imply that no standalone machine learning models have extra capability to generalize cross-dataset intrusion detection while evaluating on benchmark datasets considering huge distribution shifts and class imbalance in the datasets.

4.1 Hypothesis Analysis

In addition to the results obtained after training the models to find which model shows greater capabilities in cross-dataset generalization for NIDS, we also conducted a 10 fold significance test (p-value test) for both supervised and unsupervised machine learning models to check whether our hypotheses were significant or not. The results showed that 5 of our hypotheses were significant. H5, H7 and H8 although cannot be determined by a significance test, based on the descriptive analysis (mean cross F1_Score) we have considered H5 to be significant. However, H7 and H8 requires a predeclared feature intervention and mediation design; thus remains a further analysis.

Table 13: Hypothesis Results

S.No	Hypothesis	Significance Test (p-value)	Remarks
H1	The evaluation metrics Precision, Recall, F1_Score and ROC-AUC for intrusion detection depend significantly on the type of model used.	min Holm $p = 0.35156$	Significant
H2	The evaluation metrics of the models for intrusion detection depend significantly on the assessment setting used for evaluation.	min Holm $p = 0.35156$	Significant
H3	Attack type signifies the performance of the models in terms of evaluation metrics.	min Holm $p = 0.35156$	Significant
H4	Cross-generalization for network intrusion detection effectiveness relies on the interaction between both the selection of models and evaluation settings.	min Holm $p = 0.35156$	Significant
H5	Robustness is not determined solely by the highest score on one transfer path.	Not a p -value hypothesis	Results based on mean, worst path, and variability across all cross-dataset paths
H6	Evaluation setting and attack type have significance in generalization for intrusion detection.	min Holm $p = 0.35156$	Significant
H7	Dataset features have an intermediary role between the individual variables and dependent variables for the effectiveness of intrusion detection.	Not estimable	Requires a predeclared feature intervention and mediation design.
H8	Feature selection/type mediates the link between the model and the performance of intrusion detection.	Not estimable	Requires a predeclared feature intervention and mediation design.

Table 14: Cross-path robustness across models (Descriptive Dtatistics for H5).

Model	mean_cross_auc	worst_cross_auc	sd_cross_auc	mean_cross_f1
Supervised Models				
XGBoost	0.5961	0.1872	0.1982	0.0883
Deep Neural Network	0.5806	0.1367	0.1986	0.1517
Random Forest	0.5755	0.1401	0.2284	0.0561
Unsupervised Models				
Improved Autoencoder	0.6470	0.2644	0.2012	0.2537
One-Class SVM	0.6074	0.2708	0.2173	0.1780
Isolation Forest	0.5975	0.2503	0.2251	0.2294

Chapter 5

Discussion and Conclusion

This chapter interprets the findings relative to the research questions and objectives. The dissertation evaluated the performance of Supervised Machine Learning Models (Random Forest, XGBoost, DNN) and Unsupervised Machine Learning Models (Isolation Forest, One-Class SVM, and improved Auto Encoder) on CICIDS2017, CSE-CIC-IDS2018, and UNSW-NB15 for cross-dataset generalization for NIDS.

Two aspects of performance must be distinguished. ROC-AUC measures how well continuous anomaly scores rank attacks above benign flows across thresholds. F1_score and False-Positive Rate (FPR) describe the detector’s behaviour at the threshold selected using source validation data. A model can retain useful score discrimination after transfer while producing too many false alerts at that frozen thresholds. The implication, contribution, limitations, and conclusions are considered with this distinction in mind.

5.1 Implications

Implications for Cross-Dataset Generalization

The within-dataset results provided useful baselines but did not predict reliable performance on external datasets. We performed within dataset generalization using Supervised Models only for CICIDS2017. All the models generated F1_Score and ROC-AUC higher than 0.99; XGBoost being the best model. Isolation Forest achieved the highest within-dataset F1_Score on CICIDS2017 (0.7250) and CSE-CIC-IDS2018 (0.5700), while the improved autoencoder led on UNSW-NB15 (0.8112). Meanwhile within-dataset ROC-AUC was 0.9097 for Isolation Forest, 0.7914 for One-Class SVM, and 0.8335 for the autoencoder. The corresponding cross-dataset means were 0.6250, 0.6211, and 0.6490. These are arithmetic averages across different paths, not estimates of performance variation across repeated runs. All the models (Supervised and Unsupervised) had lower mean ROC-AUC and F1_Score across the cross-dataset paths than across their within-dataset paths.

They nevertheless describe a consistent deterioration in the recorded experiment. The results reinforces the need for external testing highlighted by [Verkerken et al., 2022] and the broader cross-dataset concerns discussed by [Cantone et al., 2024]. Transfer direction also mattered. A model trained on CICIDS2017 and evaluated on CSE-CIC-IDS2018 did not necessarily behave like a model trained on CSE-CIC-IDS2018 and evaluated on CICIDS2017. Reversing the direction changes the normal training distribution, fitted preprocessing parameters, selected configuration, and source-calibrated threshold. Cross-dataset findings should therefore name both the source and target rather than treat a dataset pair as one undirected experiment.

Implications for Intrusion Detection Practice

The findings from the comparative analysis illustrate that 'Accuracy' or 'ROC-AUC' doesn't attribute as the effective indicators of the performance of the models. Although some models showed significant ROC-AUC scores, their operational thresholds and performance were limited in most cases. For example, Isolation Forest trained on UNSW-NB15 and tested on CICIDS2017 yielded a ROC-AUC score of 0.8029, yet the threshold calibration gave a false positive rate of 0.9307, meaning that it wrongly flagged 59786 out of 64240 legitimate flows. Similarly, the Improved Autoencoder trained on CSE-CIC-IDS2018 yielded the highest ROC-AUC of 0.8279 among the three unsupervised models studied even though it had a false positive rate of 0.8866 when evaluated using CICIDS2017. Conversely, all 3 unsupervised models achieved ROC-AUC scores below 0.50, implying ineffective transfer from CSE-CIC-IDS2018 to UNSW-NB15. These results, indicate that a good discrimination indicator does not necessarily mean a good operating point for actual use.

This type of inconsistency between overall performance and successful attack detection was observed in the performance of supervised models as well. Random Forest saw accuracies of 0.9188 and 0.9743 for CSE-CIC-IDS2018 and UNSW-NB15 respectively, but its attack recall were only 0.2043 and 0.0002 respectively. XGBoost saw ROC-AUC scores of 0.9153 and 0.7829 across the two test datasets. Despite better ROC-AUC scores, the recall score stood low at 0.3150 and 0.0739 implying the model's effectiveness in detecting minority attack classes.

The outcome of DNN serves as an additional proof that the configuration of a model lead to different results across datasets. The first Optuna study of DNN achieved ROC-AUC of 0.5954 and 0.8096 with recall score of 0.0005 and 0.0794 for the 2 test datasets. The second Optuna study showed lower scores than the first one. Despite some improvements in metrics, DNN still showed inconsistent performance lacking the capability to detect attacks even after optimizing the hyperparameters.

Altogether, the results suggest that NIDS evaluation across datasets should provide information on a number of supporting metrics instead of relying on a single metric. This is significant since accuracy can still be high even if the models detect few attacks when the datasets are highly imbalanced. Despite the usefulness of ROC-AUC for evaluating ranking performance, it does not indicate whether the threshold set allows for the desired number of missings and false alarms.

None of the models showed robustness in generalizing for NIDS. Hence, they are not ready for deployments. A high false-positive rate can be very demanding for security analysts in a real intrusion detection process, while a low recall may allow attacks to remain undetected. Hence, it is important to have a clearly determined false alarm budget, the correct calibration of thresholds on representative validation data, further validation using real traffic of the target environment and an assessment of resources required for alerts investigations in order to deploy the model in real scenario.

To sum up, from the results, it can be concluded that predictive performance based on a dataset doesn't guarantee that the model can be used on other unseen datasets. Even though, if the model performs well on one benchmark dataset, it doesn't mean that the model is trustworthy in generating better results from other dataset. Thus, the model selection criteria should consider classical aggregate performance criteria as well as attack-class recall, false-positive rate, consistent performance across datasets and the operational threshold stability.

5.2 Research Contributions

The primary contribution is consistent directional comparison of three unsupervised anomaly-detection models across three intrusion-detection datasets. The experiment used a common nine-feature flow representation, source-fitted preprocessing, normal-only model training, labelled source-domain validation for model and threshold selection, and fixed target test partitions. This design made the nine source-to-target paths comparable within the recorded experiment.

A second contribution is the joint interpretation of ranking and operating-point performance. The difference between ROC-AUC and F1_Score or FPR is not a technical detail: it changes the conclusion that can reasonably be drawn about transferred detector. Several models retained some ranking ability while producing many false alerts at their source-calibrated thresholds. Reporting all three measures prevents a favourable ROC-AUC from concealing poor alert quality.

A third contribution is the demonstration that no model led every transfer path. XGBoost generated the highest ROC-AUC of 0.9153 and F1_Score of 0.4768 for CSE-CIC-IDS2018. DNN produced the highest ROC-AUC of 0.8107, best F1_Score of 0.1328 by XGBoost for UNSW-NB15 amongst supervised models. The autoencoder had the highest mean cross-dataset ROC-AUC (0.6490), while Isolation Forest had the highest mean cross-dataset F1_Score (0.3084) amongst unsupervised models. Neither model can be called uniformly robust on that basis: the difference were not tested across repeated runs, and some loading paths had high FPR.

The study does not claim a new anomaly-detection architecture or overall superiority over published methods. For the CICIDS2017-¿CSE-CIC-IDS2018 directions also evaluated by [Verkerken et al., 2022], the present results showed higher ROC-AUC for all the models but lower F1_Score. It could be due to study variation as studies differ in feature construction, sampling, pre-processing, class composition, and threshold selection, these figures do not form a controlled head-to-head test. The contribution is the transparent account of what transferred, what failed, and how the answer changed with the metric and direction.

The Research questions can be answered as follows:

- R01.** All the supervised and unsupervised machine learning models generalised unevenly to unseen datasets. Some paths retained moderate score discrimination, whereas others had poor ROC-AUC or excessive false alarms.
- R02.** Supervised Machine Learning Models outperformed Unsupervised Machine Learning Models in terms of performance for cross-dataset generalization for NIDS with higher F1_Score and ROC-AUC.
- R03.** No single model demonstrated consistently superior robustness across the dataset distribution shift in real-world network intrusion detection scenarios. XGBoost showed superior performance with F1_Score of 0.4768 and ROC-AUC score of 0.9153 while tested on CSE-CIC-IDS2018. It remained superior supervised machine learning model for UNSW-NB15 as well, however, Improved Auto-Encoder showed robustness with F1_Score of 0.3029 and ROC-AUC score of 0.7573 while tested on UNSW-NB15.

5.3 Research Philosophy

The study adopted a positivist, deductive, quantitative, and comparative approach. Model type, source dataset, target dataset, anomaly scores, predictions, and performance measures were recorded through defined computational procedure. The conclusions are based on those observations rather than participant opinions.

The deductive element involved examining expectations from prior work, that performance would often deteriorate under dataset shift, that model behavior would differ, and that transfer direction would matter. The recorded results were consistent with these expectations descriptively. However, one result per model and path does not support formal inferential testing.

This approach also places a boundary on explanation. The experiment measured changes in performance but did not isolate the cause of each change. Differences in normal traffic, attack composition, features extraction, prevalence, and threshold suitability are plausible explanations. Their individual effects cannot be established from the reported results alone.

5.4 Constraints and Limitations

- The three public benchmarks contained particular mixtures of traffic and attacks. They cannot and do not represent every current network or future attack. Dataset coverage and realism are recognised limitations of intrusion-detection. [Ring et al., 2019, Kenyon et al., 2020].
- Harmonisation required the study to use nine network-flow features that could be constructed across all three datasets. This enabled direct transfer but excluded other potentially informative fields. Similar features names also do not guarantee identical measurement processes or distribution across data sources.
- The within-dataset partitions were formed through stratified random sampling of flows. The experiment did not establish that duplicate or highly similar flows were separated, nor did it evaluate a later time period as a temporal holdout. Within-dataset performance may therefore be more favourable than performance under a stricter deployment-oriented split.
- Computational caps limited training, validation, and testing to at most 100,000 normal training flows, 50,000 validation flows, and 80,000 test flows per normal flows. The resulting estimates describe these sampled partitions, not every available flow.
- Although model fitting used normal source traffic labelled source validation data were used for model selection and threshold calibration. The procedure should not be presented as entirely label-free. The same source-validation partition supported multiple development decisions, potentially making validation performance optimistic. Target labels were not used for those decisions and remained reserved for final evaluation.
- Binary benign versus attack evaluation concealed differences among attacks categories. The experiment did not measure real-time latency, resource consumption, analyst workload, or detector behavior under continuously changing production attacks.

5.5 Scope for Future Researchers

The most immediate extension is to repeat the experiment for all the models across several independent random seeds. Reporting path-specific means, dispersion, and uncertainty intervals would allow a stronger assessment of whether the observed difference are stable. A second extension is to use deduplicated and temporal partitions, followed by testing on traffic from operationally distinct environments. These designs would provide a more demanding test of whether within-dataset performance and cross-dataset ranking persist outside random flow-level splits. Third, future studies could compare several documented common feature sets while preserving source-only statistical fitting. This would help distinguish information lost through nine-feature harmonisation from limitations of the anomaly-detection models. Fourth, threshold calibration should be evaluated explicitly. Researchers could compare source-only threshold rules and report recall at predeclared false-positive-rate budget. Unlabelled target adaptation may also be investigated, but it must be presented as a separate protocol. Target test labels must not determine model selection or thresholds. Finally, future evaluations could add precision–recall curves, average precision, attack-category analyses, and computational measurements. These would clarify the effects of class composition and the practical cost of operating each detector.

5.6 Summary

This dissertation tested whether Supervised and Unsupervised Machine learning models trained on normal source traffic dataset CICIDS2017 generalise across benchmark test datasets CSE-CIC-IDS2018 and UNSW-NB15. Within-dataset performance did not reliably carry over to external datasets, and the direction of transfer mattered. No model showed robust generalization capability for NIDS.

The central conclusion is that cross-dataset generalisation requires more than a favourable score on one benchmark or metric. ROC-AUC describes transferred score discrimination. F1_Score show the consequences of making decisions at frozen source-calibrated threshold. Several models retained some discrimination but generated excessive false alerts. The work provides a reproducible evaluation framework and a bounded account of its findings, while identifying the validation needed before claims of reliable real-world intrusion detection can be made.

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